Political Affiliation Detection in French Electoral Manifestos: A Comparative Study of TF-IDF and CamemBERT

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Abstract

We study the task of political affiliation detection on the Archelec corpus, a large collection of 33,000 digitized French electoral manifestos spanning legislative elections from 1958 to 1993. After filtering for OCR quality and normalizing political labels into six families, we obtain 15,300 labeled documents. We benchmark a TF-IDF with Logistic Regression baseline against a fine-tuned CamemBERT model. To disentangle genuine rhetorical signals from lexical shortcuts, we design a masking experiment removing explicit party and leader names. Results show that while TF-IDF achieves higher overall accuracy (94% vs 91%), CamemBERT proves significantly more robust under masking (90% vs 87%), suggesting it captures deeper stylistic patterns beyond surface lexical cues.

1 Introduction

Political manifestos are among the richest sources of political discourse. Distributed to voters before each election, they encapsulate a candidate's ideological positioning, rhetorical strategies, and policy priorities. Despite their political significance, they remain largely understudied from a computational linguistics perspective, particularly in the French context.

In this work, we address the task of political affiliation detection: given the text of a French electoral manifesto, can we automatically predict the political family of its author? We experiment on the Archelec corpus, a large collection of digitized French electoral manifestos spanning legislative elections from 1958 to 1993.

We make the following contributions:

- We construct a cleaned, labeled dataset of 15,300 manifestos across six political families from the Archelec corpus.
- We benchmark two approaches: a classical TF-IDF with Logistic Regression baseline and a fine-tuned CamemBERT model.
- We design a masking experiment to disentangle lexical cues (party names, leader names) from genuine rhetorical signals, and show that CamemBERT is significantly more robust to this masking than TF-IDF.
- We provide an interpretability analysis of the most discriminative features learned by each model.

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2 Data

2.1 The Archelec Corpus

The Archelec corpus is a digitized collection of French electoral manifestos (professions de foi) hosted by Sciences Po and archived on the Internet Archive. It covers legislative, presidential, and European elections from 1958 to 2014. Each document is associated with rich metadata including the candidate's name, profession, age, gender, political affiliation (titulaire-soutien), constituency, and election year.

We restrict our analysis to legislative elections, which represent 99.6% of the corpus, yielding a more homogeneous document collection. OCR-processed text files are available for each manifesto via dedicated URLs. We retrieved 29,444 texts out of 33,031 documents (89% retrieval rate), with failures attributable to degraded scans on the Internet Archive.

2.2 OCR Quality Filtering

OCR quality varies significantly across documents due to the age and physical condition of the original scans. We define a quality score as the ratio of tokens whose length falls between 2 and 20 characters, serving as a proxy for readable words. Documents with a quality score below 0.6 were discarded, removing 165 documents. The remaining corpus has a mean quality score of 0.91.

2.3 Political Family Labeling

The raw `titulaire-soutien` field contains over 200 distinct values, reflecting the fragmentation of the French political landscape across decades. We manually mapped these values to six political families: Gauche radicale, Gauche, Écologie, Centre-droite, Droite, and Extrême droite.

This mapping required careful historical reasoning. We treat the SFIO (Section Française de l'Internationale Ouvrière) and the Parti Socialiste as belonging to the same social-democratic family, despite being formally distinct organizations (the PS replaced the SFIO in 1969). Similarly, electoral coalitions are assigned to a family only when all constituent parties belong to the same group; ambiguous multi-party coalitions are discarded.

Documents with no mappable affiliation (non mentionné, indépendant, sans étiquette) were excluded, as they provide no reliable training signal. This yields a final dataset of 15,300 labeled documents.

2.4 Dataset Statistics

The final dataset covers elections from 1958 to 1993, with the bulk concentrated in legislative elections of 1958, 1962, 1967, 1968, 1973, 1978, 1981, and 1988. The class distribution is imbalanced, with Gauche radicale being the most represented family (6,579 documents) and Écologie the least (427 documents). The mean document length is 5,380 characters (std: 2,541), with a minimum of 200 and a maximum of 29,857 characters. Regarding gender, 89.5% of candidates are male, reflecting the historical underrepresentation of women in French politics during this period.

Table 1: Dataset statistics by political family.

Family	Documents	% of corpus
Gauche radicale	6,579	43.0
Gauche	3,253	21.3
Centre-droite	2,590	16.9
Droite	1,395	9.1
Extrême droite	1,056	6.9
Écologie	427	2.8
Total	15,300	100

Figure 1 shows the distribution of documents across election years. The corpus is concentrated around the major legislative elections of the Fifth Republic, with peaks in 1973 (2,172 documents)

and 1978 (2,412 documents). A small number of documents from post-2000 elections are present but excluded from the analysis due to insufficient representation.

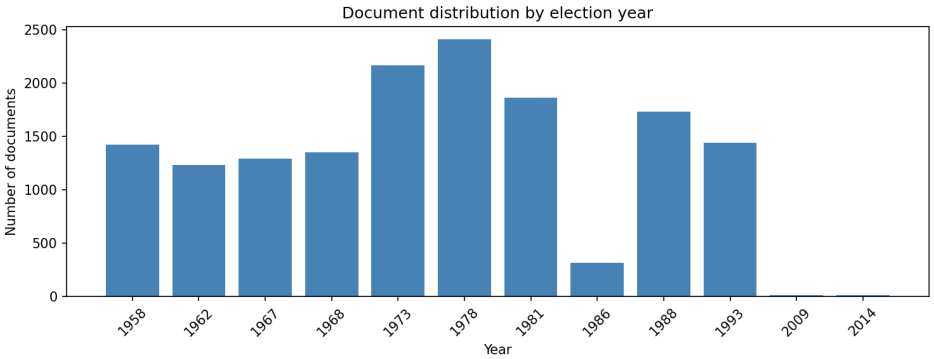

Figure 1: Distribution of documents by election year.

3 Related Work

The analysis of political texts using NLP has gained significant attention over the past decade. Early work by Laver et al. [1] introduced word-score methods to extract policy positions from party manifestos, establishing the foundation for computational political science. The Comparative Manifestos Project has since provided large-scale annotated corpora enabling supervised approaches [2].

On the classification side, Iyyer et al. [3] pioneered the use of recursive neural networks for political ideology detection, demonstrating that syntactic structure carries ideological signal beyond bag-of-words features. More recently, Nikolaev et al. [4] compared label aggregation pipelines against long-input Transformer models on the CMP dataset across 41 countries and 27 languages, finding that state-of-the-art models can effectively solve left-right scaling tasks. Orellana and Bisgin [5] applied BERT embeddings to analyze New Zealand party manifestos, combining document similarity, topic modeling, and sentiment analysis.

The comparison between classical and neural approaches for text classification has been extensively studied. Gonzalez-Carvajal and Garrido-Merch [6] empirically showed that BERT does not systematically outperform TF-IDF across all tasks and dataset sizes, a finding consistent with our results.

Our work differs from prior studies in three key aspects: we focus on French-language manifestos spanning several decades, we introduce a controlled masking experiment to isolate rhetorical signals from lexical shortcuts, and we use CamemBERT [7], a language model specifically pre-trained on French text.

4 Methods

4.1 Baseline: TF-IDF with Logistic Regression

Our baseline represents each manifesto as a TF-IDF vector computed over a vocabulary of 10,000 unigrams and bigrams, with a minimum document frequency of 3. We then train a Logistic Regression classifier with L2 regularization ($C = 1.0$) and `class_weight='balanced'` to account for class imbalance. The model is trained on 80% of the data and evaluated on the remaining 20%, with stratified splits to preserve class proportions.

TF-IDF was chosen as a baseline for three reasons: it is computationally efficient, highly interpretable through its feature weights, and has been shown to perform competitively against more complex models on document classification tasks [6].

4.2 Fine-tuned CamemBERT

Our neural approach fine-tunes CamemBERT-base [7], a RoBERTa-based model pre-trained on 138GB of French text. We add a linear classification head on top of the `[CLS]` token representation and fine-tune all parameters for 3 epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and a batch size of 16. Input texts are truncated to 512 tokens, corresponding to the first 1,500 characters of each manifesto. Class imbalance is handled via a weighted cross-entropy loss, with weights inversely proportional to class frequencies. CamemBERT experiments were run on a Google Colab T4 GPU.

4.3 Masking Experiment

To assess whether models rely on genuine rhetorical signals or merely on lexical shortcuts, we design a masking experiment. We compile a list of 47 terms including party names (e.g., *parti communiste*, *front national*), abbreviations (e.g., *udf*, *pcf*), and political leader names (e.g., *de gaulle*, *mitterrand*). These terms are removed from all documents using whole-word regex matching before vectorization or tokenization. Both models are then retrained and evaluated on this masked corpus under identical conditions.

4.4 Interpretability Analysis

For the TF-IDF model, we extract the top 15 features per class by ranking the Logistic Regression coefficients. This analysis is performed on both the original and masked corpora, allowing us to contrast what each model learns with and without explicit political cues.

5 Results

5.1 Classification Performance

Table 2 reports accuracy and macro F1 scores for all four experimental conditions.

Table 2: Classification performance across models and corpus conditions.

Model	Accuracy	Macro F1
TF-IDF + LR (original)	0.94	0.93
CamemBERT (original)	0.91	0.89
TF-IDF + LR (masked)	0.87	0.84
CamemBERT (masked)	0.90	0.87

In the original corpus, TF-IDF outperforms CamemBERT by 3 points in accuracy. However, after masking, TF-IDF drops by 7 points while CamemBERT drops by only 1 point, nearly closing the gap between the two models.

5.2 Per-class Analysis

Both models achieve their highest F1 scores on Gauche radicale (0.97–0.98) and Extrême droite (0.88–0.97), reflecting the highly distinctive vocabulary of these families. The most difficult class is Centre-droite, which is systematically confused with Droite, consistent with their rhetorical proximity in the t-SNE projection.

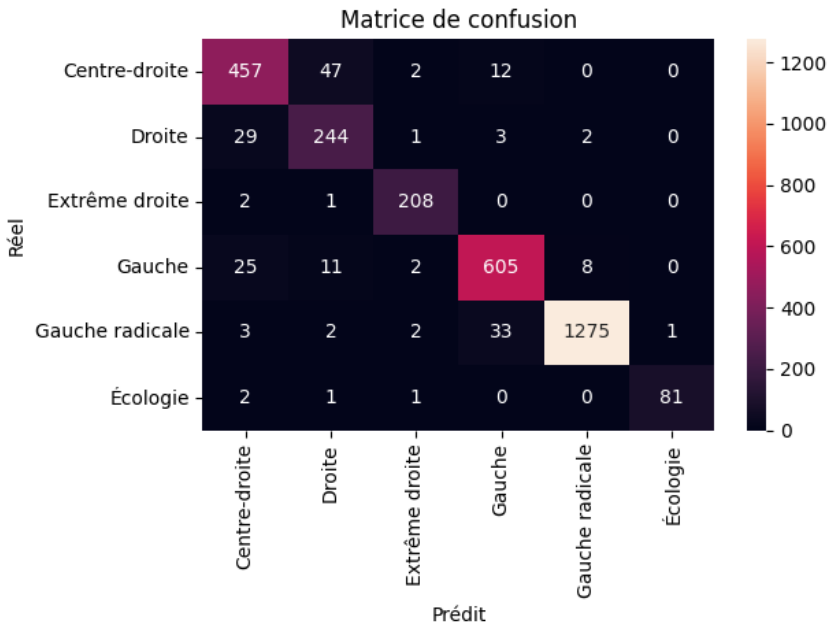

Figure 2: Confusion matrix of the TF-IDF + LR model on the original (unmasked) test set.

5.3 Geometric Analysis

Figure 3 shows the t-SNE projection of TF-IDF representations for a balanced sample of 1,200 documents. Extrême droite and Écologie form tight, well-separated clusters, while Droite and Centre-droite overlap significantly in the center of the embedding space. Gauche radicale occupies a distinct region to the left, whereas Gauche documents form a transition zone between Gauche radicale and the centrist families.

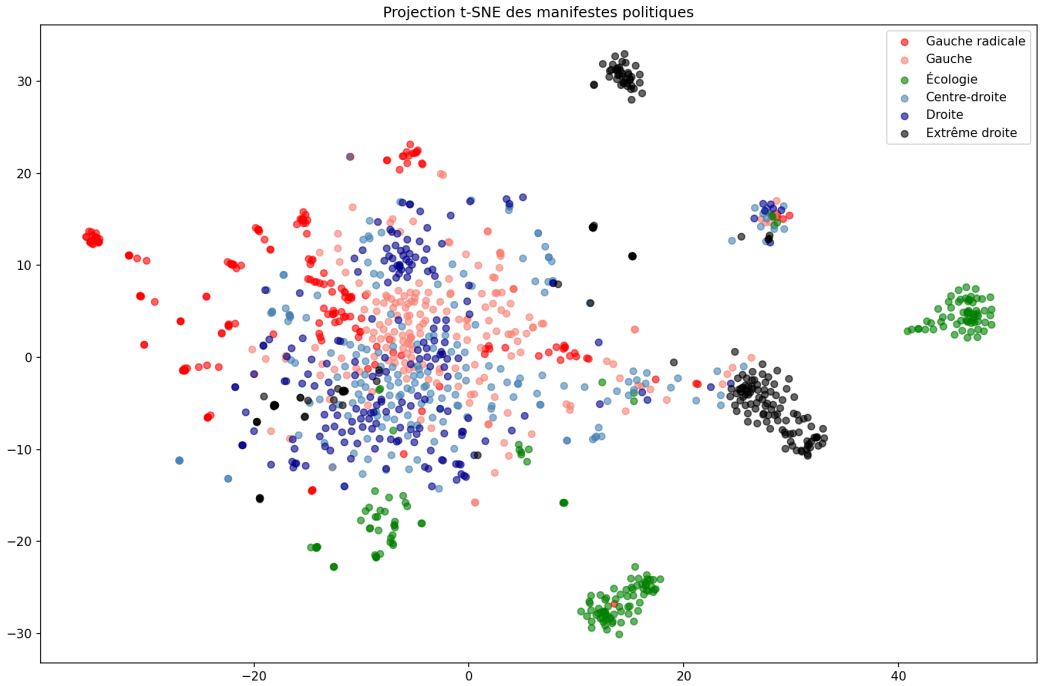

Figure 3: t-SNE projection of TF-IDF document representations, colored by political family.

5.4 Interpretability

Figure 4 displays the top 15 discriminative features per class from the masked TF-IDF model. Without explicit party names, the model recovers politically coherent vocabulary: Extrême droite is characterized by *immigration*, *insécurité*, and *immigrés*; Gauche radicale by *travailleurs*, *ouvrier*, and *forces*; Écologie by *environnement*, *vivante*, and *génération*. Droite retains strong gaulliste markers such as *le général* and *union*, despite the removal of leader names.

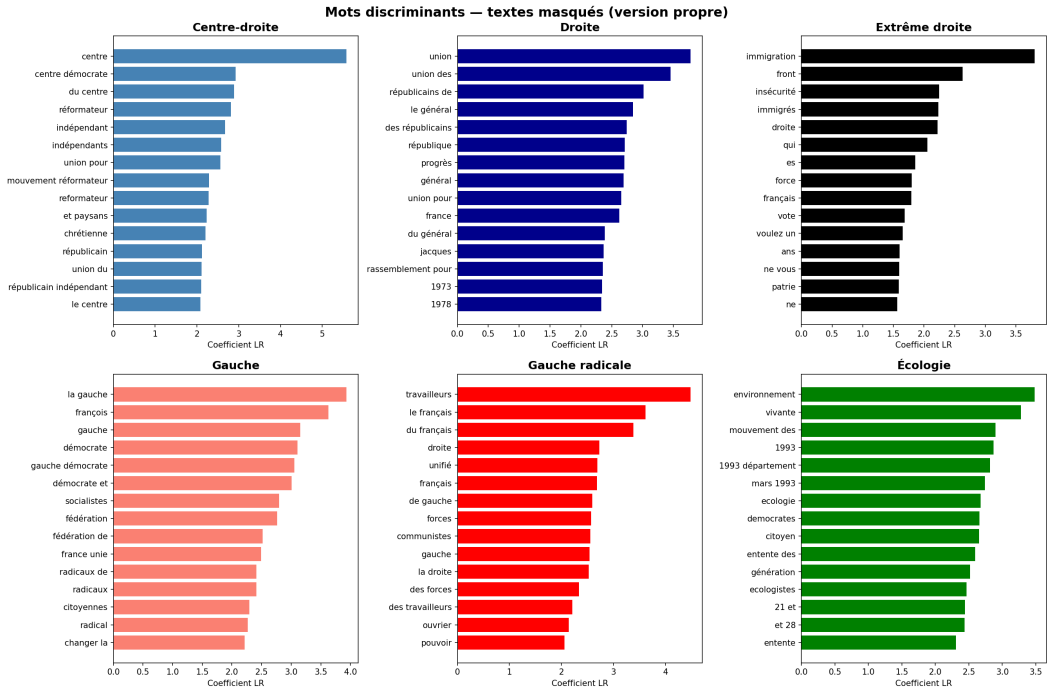

Figure 4: Top 15 discriminative features per political family (masked corpus).

6 Discussion and Conclusion

This work demonstrates that political affiliation can be detected with high accuracy from the text of French electoral manifestos. The most important finding is not the raw performance of either model, but what the masking experiment reveals about the nature of the signal each model captures.

TF-IDF relies heavily on explicit lexical cues: its accuracy drops by 7 points when party names and leader names are removed. This suggests that a significant portion of its performance stems from direct mentions of political entities rather than from stylistic or thematic patterns. CamemBERT, by contrast, loses only 1 point under masking, indicating that its contextual representations encode deeper rhetorical patterns that persist even in the absence of surface-level identifiers.

The confusion matrix analysis reinforces a politically coherent picture: the most frequent misclassifications occur between Droite and Centre-droite, two families that share significant rhetorical overlap in the French political landscape. Conversely, Extrême droite and Gauche radicale are almost perfectly classified, reflecting the highly distinctive and stable vocabulary of these families across decades.

The interpretability analysis on masked texts reveals that models learn historically grounded features. Gauche radicale manifests through class-based vocabulary (*travailleurs*, *ouvrier*), Extrême droite through nativist rhetoric (*immigration*, *insécurité*), and Écologie through environmental and civic lexicon (*environnement*, *génération*). These findings validate the political coherence of our family groupings and confirm that the models capture genuine ideological signals.

Limitations. Several limitations should be acknowledged. First, our political family mapping involves subjective historical judgments, particularly regarding parties that evolved significantly over the study period. Second, OCR noise in older documents may systematically affect certain elections more than others. Third, truncating documents to 1,500 characters for CamemBERT discards potentially informative content from longer manifestos. Finally, we evaluate on a single train-test split; results may vary across different random seeds.

Future work. Promising extensions include applying token-level attribution methods such as Integrated Gradients or LIME [8, 9] to identify which specific passages drive CamemBERT's predictions, evaluating generalization to more recent elections from the Regards Citoyens corpus, and exploring temporal drift in political rhetoric across the 35-year span of the corpus.

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